

```
In [1]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from sklearn.linear_model import LinearRegression
from sklearn.model_selection import train_test_split
from sklearn.metrics import r2_score
```

```
In [2]: df=pd.read_csv("house_price.csv")
df.head()
```

```
Out[2]:
```

	House Size	Bedrooms	Age of House	House Price
0	1500	3	10	350000
1	1800	3	8	400000
2	2000	4	5	450000
3	2200	4	2	475000
4	2500	5	15	525000

```
In [3]: x = df[['House Size', 'Bedrooms', 'Age of House']]
y = df['House Price']
```

```
In [4]: X_train,X_test,Y_train,Y_test=train_test_split(x,y,test_size=0.2,random_state=42)
```

```
In [5]: model=LinearRegression()
model.fit(X_train,Y_train)
```

```
Out[5]:
```

▼ LinearRegression ⓘ ?

► Parameters

```
In [6]: Y_pred=model.predict(X_test)
Y_pred
```

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Out[6]: array([ 958198.91326438, 3084757.95178209, 2213923.0591368 ,
                3676539.75229267, 1145827.56632132, 3989268.46066852,
                1827657.81346492, 1706685.73548026, 2419439.6139755 ,
                1275055.59897449])
```

```
In [7]: r2_score=r2_score(Y_test,Y_pred)
        r2_score
        #print evaluation metrics
        print(f"R-squared Score:",r2_score)
```

R-squared Score: 0.9987416757120748

```
In [8]: #plot
        plt.scatter(x['House Size'],y,color='maroon',label='Data points') #All data points
        plt.plot(x['House Size'],model.predict(x),color='red',label='Regression Line') #Regression Line
        plt.xlabel('House Size(sqft)')
        plt.ylabel('House Price($)')
        plt.title('Multiple Linear Regression')
        plt.legend()
        plt.show()
```

